

MTH209: Worksheet 12

Bayes Classifier

Suppose observe data (x_i, y_i) , $x_i \in \mathbb{R}^d$, $y_i \in \{1, \dots, K\}$. This is a supervised setup, since labels are given. The goal is that given a new point x_{new} , predict its class. Naturally, we would assume that each class label k has a different behaving population:

$$x_i \mid y_i = k \sim F_k.$$

Further, we acknowledge the fact that different classes may have different proportion in the population. So

$$P(y = k) = \pi_k.$$

Using this setup, and the Bayes theorem, we can arrive at the conditional distribution of an observation having label k given the data observed.

$$P(y_i = k \mid x_i) \propto \pi_k f_k(x_i \mid y_i = k)$$

Now, using this, one can setup the follow estimator of the class label of x_i , with the intuition that we want that class label that has the highest probability:

$$\hat{y}_i(x_i) = \arg \max_k \{ \pi_k f_k(x_i \mid y_i = k) \}$$

This is called the Bayes Classifier method, since it employs the Bayes theorem to obtain a predicted class label. Different assumptions on F_k lead to different methods of classification. One of them is *Linear Discriminant Analysis*.

Linear Discriminant Analysis

In this method, we assume that F_k is Gaussian, but specifically, for $k = 1, \dots, K$:

$$x_i \mid y_i = k \sim N_p(\mu_k, \Sigma)$$

Note that the interpretation of this is that all means of the K groups are different, but the covariance structure is the same. This yields the pdf

$$f_k(x \mid y = k) \propto \exp \left(-\frac{1}{2} (x - \mu_k)^\top \Sigma^{-1} (x - \mu_k) \right)$$

If we expand the exponential term, we obtain

$$(x - \mu_k)^T \Sigma^{-1} (x - \mu_k) = x^T \Sigma^{-1} x - 2\mu_k^T \Sigma^{-1} x + \mu_k^T \Sigma^{-1} \mu_k.$$

But note that the term $x^T \Sigma^{-1} x$ does not depend on k , so it will not factor into the Bayes classifier optimization. So the Bayes classifier maximizes (converting to log scale)

$$\begin{aligned} \hat{y}_i(x_i) &= \arg \max_k \left\{ \log \pi_k - \frac{1}{2} (x_i - \mu_k)^T \Sigma^{-1} (x_i - \mu_k) \right\} \\ \hat{y}_i(x_i) &= \arg \max_k \underbrace{\left\{ \log \pi_k + \mu_k^T \Sigma^{-1} x_i - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k \right\}}_{\delta_k(x_i)} \end{aligned}$$

Here, $\delta_k(x_i)$ is called the discriminant, since this quantity will allow us to “discriminate” between the classes. Great! Now that we have setup this problem, we now just need to solve this optimization problem. Notice, the maximization is over a discrete parameter, so we can’t take derivatives etc. Instead, we can solve this optimization problem using logic!

Decision Boundary

Suppose, for now, $k = 2$. Then we just need to figure out when $\delta_1(x) > \delta_2(x)$ and when $\delta_1(x) < \delta_2(x)$. This means, we should first figure out when they are equal; the point of equality will give us a decision boundary:

$$\delta_1(x) = \delta_2(x)$$

This yields that:

$$(\mu_1 - \mu_2)^T \Sigma^{-1} x - \frac{1}{2} (\mu_1^T \Sigma^{-1} \mu_1 - \mu_2^T \Sigma^{-1} \mu_2) + \log \frac{\pi_1}{\pi_2} = 0$$

This is the boundary! So if the $LHS > 0$, $\delta_1(x)$ wins and if $LHS < 0$, $\delta_2(x)$ wins! The above boundary, note, is the equation of a “line” or a plane. Further, the class probabilities, π_k shift the boundary line, not the slope.

This gives us the tools to assign the label! For larger k , all pairwise decision boundaries can be calculated and classification happens based on all the boundaries. All of this is at the population level. Given the data, we will then first estimate π_k s, μ_k s and Σ , and then use these estimates to classify.

Estimation

The estimation of μ_k s is simple. It is just the sample mean of the x s that are in the k th class label:

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} x_i$$

where n_k is the number of observations with the k th class label. Further, π_k s are also straightforward for estimation:

$$\hat{\pi}_k = \frac{n_k}{n}.$$

The estimator of Σ is slightly different, since we assume all k populations have the same Σ . Thus, Σ is calculated from all the observations together. We use the fact that $x_i|y_i = k \sim N(\mu_k, \Sigma) \Rightarrow (x_i - \mu_k)|y_i = k \sim N(0, \Sigma)$. Thus, centering each observation around its own mean, and then calculating the sample covariance matrix is the way to go. However, since K means are being estimated, we divide by $n - K$:

$$\hat{\Sigma} = \frac{1}{n - K} \sum_{k=1}^K \sum_{i: y_i = k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T.$$

Once these parameters are estimated, we can now use these in predictions!

Questions

1. In this first exercise, we will code everything from scratch. First, let's generate data for two clusters:

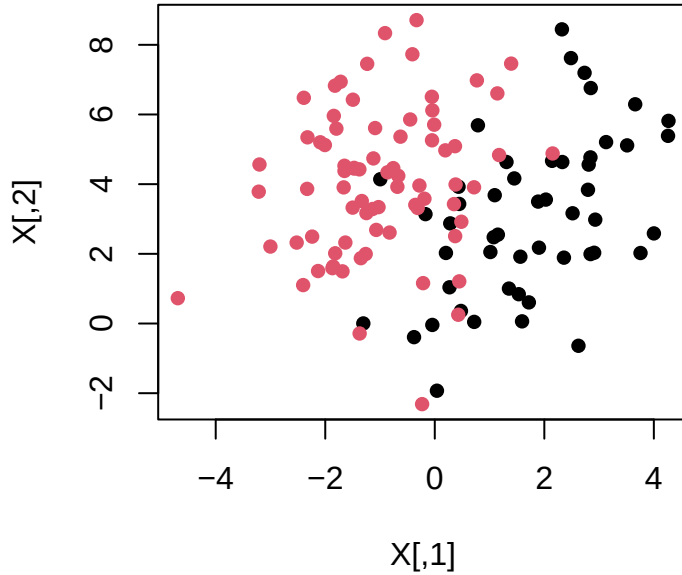
```
library(MASS) # for generating multivariate normal
set.seed(209)
Sigma <- matrix(c(2,1,1,4), nrow = 2, ncol = 2)
mu1 <- c(2, 3)
mu2 <- c(-1, 4)
n1 <- 50
n2 <- 75

# generate xs
x1 <- mvrnorm(n1, mu = mu1, Sigma = Sigma)
x2 <- mvrnorm(n2, mu = mu2, Sigma = Sigma)

# fix ys
y1 <- rep(1, n1)
y2 <- rep(2, n2)
y <- c(y1, y2)
X <- rbind(x1, x2)
```

Let us plot the above data to visualize

```
plot(X, col = y, pch = 16)
```



- a. Calculate $\hat{\pi}_k$ s, $\hat{\mu}_k$ s, and $\hat{\Sigma}$.
 - b. Write a function called `decision_boundary()` that takes an input x , and estimates to report whether $\hat{\delta}_1(x) - \hat{\delta}_2(x) > 0$ or < 0 , where:

$$\hat{\delta}_1(x) - \hat{\delta}_2(x) = (\hat{\mu}_1 - \hat{\mu}_2)^T \hat{\Sigma}^{-1} x - \frac{1}{2} (\hat{\mu}_1^T \hat{\Sigma}^{-1} \hat{\mu}_1 - \hat{\mu}_2^T \hat{\Sigma}^{-1} \hat{\mu}_2) + \log \frac{\hat{\pi}_1}{\hat{\pi}_2}$$
 - c. For each observation in the dataset, obtain $\hat{y}(x)$, and calculate the mis-classification rate.
2. We can implement LDA prediction easily in R using the `MASS` package and also visualize pair-wise boundary regions using the `klaR` R package.

- a. Load the `iris` data in R and verify if all classes have roughly the same Σ .
- b. Run LDA using the code below:

```
library(MASS)
library(klaR)
lda_fit <- lda(Species ~ ., data = iris)

# Plot pairwise decision boundaries
partimat(Species ~ ., data = iris, method = "lda")
```

Interpret the partial boundary region plot. Is the mis-classification region low or high?

- c. Using `predict()` function, calculate the mis-classification rate of this data.
3. For the B3 dataset in R, implement LDA with PHASEN being the label.
- a. Obtain the mis-classification rate and compare with that of kNN.
4. Although we have not learned Quadratic Discriminant Analysis (QDA), QDA essentially does not assume all clusters have the same Σ . The general method then is the same, except the boundary lines turn into parabolas.

- a. Implement QDA on the `iris` data:

```
lda_fit <- qda(Species ~ ., data = iris)

# Plot pairwise decision boundaries
partimat(Species ~ ., data = iris, method = "qda")
```

Calculate the misclassification rate and compare with LDA.

- b. Repeat a. for the B3 dataset.